SSC26-VIII-01

Validated Lie Group Variational Integrator-Based Simulator for Passive Magnetic Attitude Control of the GASRATS CubeSat

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ABSTRACT

Passive magnetic attitude control (PMAC) systems are selected for integration in CubeSat designs on account of their simplicity and lack of power consumption. These systems function by simultaneously dissipating a satellite's rotational energy to heat and orienting it according to the Earth's magnetic field. This process is critical for establishing steady-state attitude, a stable rotational state required for conducting mission operations and ensuring reliable downlink to ground stations. Comprehensive attitude simulations are necessary for confirming the reliability of a PMAC system to provide timely attitude convergence for a wide range of initial deployment velocities and orientations. An attitude damping phase precedes steady-state, and the duration may last multiple weeks or longer. As simulation timespans of one month or longer are needed to prove a stable convergence, standard adaptive Runge-Kutta (4,5) solvers (RK45) are unsuitable due to the accumulation of numerical error. Lie group variational integrators (LGVIs) offer a compelling alternative. LGVIs conserve angular momentum and inherently preserve attitude matrix orthogonality, resulting in improved physics accuracy. A three-step validation chain is followed to verify a developed LGVI simulation framework, comprising a validation to literature, cross-framework verification, and invariant preservation analysis. In a torque-free scenario, the LGVI simulator is found to maintain a relative angular momentum error on the order of 10^{-13} while halving the execution time of a RK45 simulator configured with a comparable timestep. The validated LGVI simulator is employed to model the performance of eight PMAC configurations for the Get Away Special Radio and Antenna Transparency Satellite in a 1,001-trial Monte Carlo analysis. The results inform the final design recommendation of a four-rod, 1.5U configuration and reveal the unexpectedly poor damping performance of 1U configurations. It is determined that this phenomenon is caused by inertial symmetry that limits the transfer of rotational energy between body axes.

INTRODUCTION

Lie Group Variational Integrators (LGVIs) are selected for implementation in the next evolution of attitude simulations for the Get Away Special Radio and Antenna Transparency Satellite (GASRATS) by virtue of their reported physics accuracy and computational efficiency in comparison with Runge-Kutta (RK) integrators. LGVIs are of interest to attitude simulations of passive magnetic attitude control (PMAC) systems because they conserve momentum and preserve the orthogonal structure of the attitude matrices. The LGVI is a modern integration method that was pioneered in the 2000s and spearheaded by Lee et al. (2005).[1] LGVIs have been used in PMAC simulations of past CubeSat missions, such as the University of Michigan's first Radio Aurora Explorer (RAX), on account of their energy preservation over long simulation timespans. As PMAC simulations for CubeSats operating in low Earth orbit (LEO) target timespans of one month or longer to prove rotational stability, RK integrators are inadequate due to the accumulation of numerical error. Although LGVIs offer substantial improvements over traditional integration methods, their considerable complexity offers a serious challenge for producing valid implementations. Existing literature regarding LGVI simulations for CubeSat missions adopts a high-level approach without providing an approachable explanation of the underlying theory or discussing the details of the simulation algorithm. On the other hand, the source literature on LGVIs is largely inaccessible to undergraduates due to the mathematical depth. The purpose of the LGVI theory presented in this paper is to provide a comprehensible guide for fellow undergraduates, thereby enabling the proliferation of LGVIs for other CubeSat teams.

A LIE GROUP VARIATIONAL INTEGRATOR

Geometric and Numerical Foundations

The desirable energy- and structure-preserving qualities of the LGVI stem from its underlying architecture. Contrasting LGVIs with standard RK integrators highlights the nuances of the variational integration method. For standard adaptive RK methods—such as the Runge-Kutta (4,5) solver (RK45)—used for attitude modeling in a quaternion framework, the state trajectory is approximated by computing fourth- and fifth-order accurate estimates of the next state at each timestep. Instead of approximating the state trajectory, the LGVI approximates the system's action, and discrete Lagrangian equations of motion are derived from a discrete action sum via Hamilton's principle. The attitude evolves natively on the manifold of valid rotation matrices, the special orthogonal group SO(3). Thus, the matrices used for attitude representation remain orthogonal. Derived from a discrete variational principle, the LGVI is symplectic and exhibits excellent long-term energy behavior.

One downside of RK45 attitude simulations that employ quaternion kinematics is norm drift, meaning that the magnitude of the unit quaternion shifts off unit during integration. Error-introducing normalization steps are required to force quaternions back to unit length, impacting the accuracy of the simulation. RK45 integrators operate by treating a quaternion as four coordinates in 4D vector space. In the integration step, the derivative at a point is calculated, and a tangential move is made that deviates off the unit quaternion hypersphere. A normalization step reprojects the non-unit quaternion back onto the unit sphere, resulting in numerical drift that affects simulation accuracy. Fig. 1 provides a geometric illustration of normalization for three integration schemes.

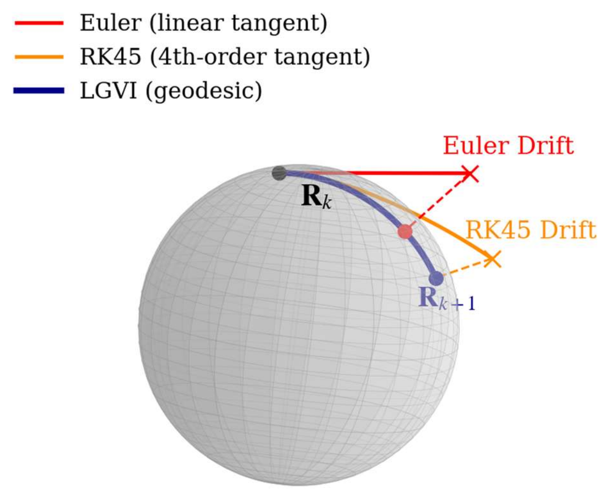

Figure 1. Position Update Mapping for Three Integration Methods Represented on the S^3 Manifold

In contrast, the LGVI method evolves a rotation matrix of SO(3) strictly on the SO(3) manifold by a multiplicative update constructed via exponential mapping. SO(3) is the group of all valid rotation matrices that satisfy the conditions $R^T R = I$ and $\det(R) = +1$. The integrator utilizes exponential mapping to generate rotation updates on the manifold, producing geodesic paths between timesteps. LGVIs for attitude modeling employ rotation matrices instead of quaternions due to the numerical issues that arise from quaternion sign duplicity, meaning that the negative and positive alternates of a quaternion represent the same physical orientation. This produces complications in the integration methods that motivate the selection of rotation matrices for attitude representation in LGVIs.

It may seem counterintuitive that LGVIs are notably computationally efficient in comparison to RK45 integrators—LGVIs use a specialized Newton iteration to solve an implicit momentum balance equation as well as heavier 3×3 rotation matrices. Conversely, researchers from the University of Michigan demonstrated that the LGVI approach offers improvements on simulation runtime in comparison to RK45. Work conducted by Lee et al. (2011) on simulations for the RAX CubeSat reported that their LGVI integration speed was 3.8 times faster than an RK45 model configured to corresponding timesteps for long duration simulations, excluding hysteresis torques.[2] For an RK45 integrator to achieve similar energy preservation to LGVIs, its timestep would have to decrease. This would increase runtime and further widen the efficiency gap between the two integrators. Thus, an LGVI is significantly more efficient than an RK45 configured to provide similar physics fidelity. The computational efficiency of the LGVI is another property of the integrator that proves its usefulness for the time-intensive Monte Carlo analysis later discussed that tests many hundreds of initial conditions.

Mathematical Formulations

The theory of the LGVI is presented in a way that directly ties to its MATLAB implementation and is based on the derivations presented by Lee et al. (2005). The purpose of this explanation is to serve as a bridge to the LGVI literature by providing an intuitive guide for undergraduates. This section highlights important steps in the derivation of the discrete Euler-Lagrange equation and the implicit momentum balance. The residual of the implicit momentum balance is provided, which forms the core of the Newton iteration. Therein, a rotation increment required for the attitude update is found that satisfies the residual equation. A systematic understanding of Hamilton's principle is essential for implementing LGVIs in MATLAB simulations and carries significant value due to its widespread occurrence throughout physics, including the action-based derivation of the Einstein field equations. By enforcing that the system follows a path such that its discrete action is stationary, the resulting integrator generates trajectories that are consistent with the physical laws governing the system's natural motion. The derivation of the discrete Euler-Lagrange equation begins with the attitude kinematic equation. The configuration space is SO(3), and the kinematic equation is given by

$$\dot{R} = RS(\Omega), \tag{1}$$

where $R \in \mathrm{SO}(3)$ is the attitude matrix and $S(\Omega)$ is the skew-symmetric matrix associated with the angular velocity $\Omega \in \mathbb{R}^3$ in the body frame.

Attitude simulations require discrete versions of such equations. Using the approximation $S(\Omega_k) = (F_k - I_{3\times 3})/h$, the discrete kinematic equation can be expressed as

$$R_{k+1} = R_k F_k, \tag{2}$$

where F_k is the rotation increment matrix. LGVIs follow an energy-based formulation and are governed by discrete equations of motion in Lagrangian form. A Lagrangian can be defined as the difference between kinetic energy T and potential energy U, and a general form for the configuration space SO(3) is given by

$$L(R, \Omega) = T(\Omega) - U(R). \tag{3}$$

For a rigid body, kinetic energy is the sum of the motion of every particle within that body, and the continuous Lagrangian equation can be expressed as

$$L(R, \Omega) = \frac{1}{2}\int_{Body} \|\Omega \times \tilde{\rho}\|^2 \, dm - U(R), \tag{4}$$

where $\tilde{\rho} \in \mathbb{R}^3$ represents a vector from the center of mass of the satellite to an individual mass element dm, and the velocity of a mass element is $\Omega \times \tilde{\rho}$.

This continuous equation is altered from vector to matrix form to provide compatibility with Lie group operations. Skew mapping replaces the cross product of Ω and $\tilde{\rho}$, and the trace identity $\|x\|^2 = tr(xx^T)$ is applied. The non-standard moment of inertia matrix J_d is preferred over the standard moment of inertia J for matrix operations.

$$J_d = \int_{Body} \tilde{\rho}\tilde{\rho}^T dm \in \mathbb{R}^{3\times 3}$$

is substituted, resulting in the continuous Lagrangian for a system evolving on SO(3), which can be expressed as

$$L(R, \Omega) = \frac{1}{2} tr[S(\Omega) J_d S(\Omega)^T] - U(R). \tag{5}$$

The discrete Lagrangian is found by approximating the action integral over one timestep with a trapezoidal rule to maintain second-order numerical accuracy. The discrete approximation of the skew-symmetric angular velocity is substituted, where h is the duration of one timestep:

$$L_d(R_k, F_k) \approx \int_{t_k}^{t_{k+1}} \frac{1}{2} tr[S(\Omega) J_d S(\Omega)^T] - U(R)\ dt,$$

$$L_d(R_k, F_k) = \frac{1}{h} tr[(I_{3\times3} - F_k)J_d] - \frac{h}{2}U(R_k) - \frac{h}{2}U(R_{k+1}). \tag{6}$$

Hamilton's principle states that the action is stationary along the physical trajectory of the system, such that its first variation is zero for a given time interval. The action is defined by the integral of (5) from initial time t_0 to final time t_f and is given by

$$\mathfrak{S} = \int_{t_0}^{t_f} \frac{1}{2} tr[S(\Omega)J_d S(\Omega)^T] - U(R)\; dt\,. \tag{7}$$

The discrete action sum is the discrete counterpart of (7). It can be expressed as

$$\mathfrak{S}_d = \sum_{k=0}^{N-1} \frac{1}{h} tr[(I_{3\times3} - F_k)J_d] - \frac{h}{2}[U(R_k) + U(R_{k+1})]. \tag{8}$$

The varied discrete action sum is found by substituting the varied rotation increment $F_k^\epsilon = e^{-\epsilon\eta_k} F_k e^{\epsilon\eta_{k+1}}$, where $\epsilon \in \mathbb{R}$ and the variation $\eta \in \mathfrak{so}(3)$. The result can be expressed as

$$\mathfrak{S}_d^\epsilon = \sum_{k=0}^{N-1} \frac{1}{h} tr[(I_{3\times3} - e^{-\epsilon\eta_k} F_k e^{\epsilon\eta_{k+1}})J_d] - \sum_{k=0}^{N-1} \left\{\frac{h}{2}U(R_k e^{\epsilon\eta_k}) + \frac{h}{2}U(R_{k+1} e^{\epsilon\eta_{k+1}})\right\}. \tag{9}$$

The first variation of discrete action is found by differentiating (9) and evaluating the perturbation parameter ϵ at zero. The boundary condition $\eta_0 = \eta_N = 0$ is applied, and the shorthand notation $U_k = U(R_k)$ for discrete potential energy is adopted. After simplifying, Hamilton's principle is invoked by setting the sum equal to zero, yielding

$$\left.\frac{d}{d\epsilon}\right|_{\epsilon=0} \mathfrak{S}_d^\epsilon = \sum_{k=1}^{N-1} tr\left[\eta_k \left\{\frac{1}{h}(F_k J_d - J_d F_{k-1}) + h R_k^T \frac{\partial U_k}{\partial R_k}\right\}\right] = 0. \tag{10}$$

Because the discrete variation η_k is a skew-symmetric matrix, the expression within the braces of (10) is symmetric. Thus, matrix identity $A - A^T = 0$ can be applied, yielding the discrete Euler-Lagrange equation:

$$\frac{1}{h}(F_{k+1}J_d - J_d F_k - J_d F_{k+1}^T + F_k^T J_d) = h\left(\frac{\partial U_{k+1}}{\partial R_{k+1}}^T R_{k+1} - R_{k+1}^T \frac{\partial U_{k+1}}{\partial R_{k+1}}\right). \tag{11}$$

(11) paired with (2) forms the discrete equations of motion in Lagrangian form. The left-hand side of (11) is the discrete kinetic variation term, representing the evolution of the system's angular momentum. The right-hand side of (11) represents the discrete torque generated by the internal potential U.

These components emerge in an equivalent relationship to characterize a discrete-time momentum balance for the system. Because F_{k+1} is formulated via exponential mapping, it remains an orthogonal matrix. This provides the structure-preserving property of the LGVI, ensuring the attitude matrix evolves strictly on the SO(3) manifold. By enforcing the stationary condition $\delta\mathfrak{S}_d = 0$ of Hamilton's principle, it is guaranteed that the resulting update map is symplectic, meaning that the geometry of the phase space is preserved. This results in excellent long-term energy stability that is advantageous for long-duration PMAC simulations.

While elegant, (11) is not practical for implementation into code because of its implicit nature and the input requirements for current and future timesteps. Applying a discrete Legendre transform introduces Π_k to represent discrete angular momentum. This results in the following equation in which $M_k \in \mathbb{R}^3$ represents the discrete torque acting on the body in Hamiltonian form:

$$S(\Pi_k) = \frac{1}{h}(F_k J_d - J_d F_k^T) - \frac{h}{2}S(M_k). \tag{12}$$

Lee et al. (2005) derive M_k from an internal gravitational potential; here, M_k is evaluated from external magnetic torques. By performing algebraic manipulations and scaling by timestep h, a transition to the implicit momentum balance is made:

$$hS\left(\Pi_k + \frac{h}{2}M_k\right) = F_k J_d - J_d F_k^T. \tag{13}$$

Whereas the residual of (13) is used to find the rotation increment F_k required for the attitude update, a discrete momentum update equation is needed to compute the angular momentum for the next timestep:

$$\Pi_{k+1} = F_k^T \Pi_k + \frac{h}{2}F_k^T M_k + \frac{h}{2}M_{k+1}. \tag{14}$$

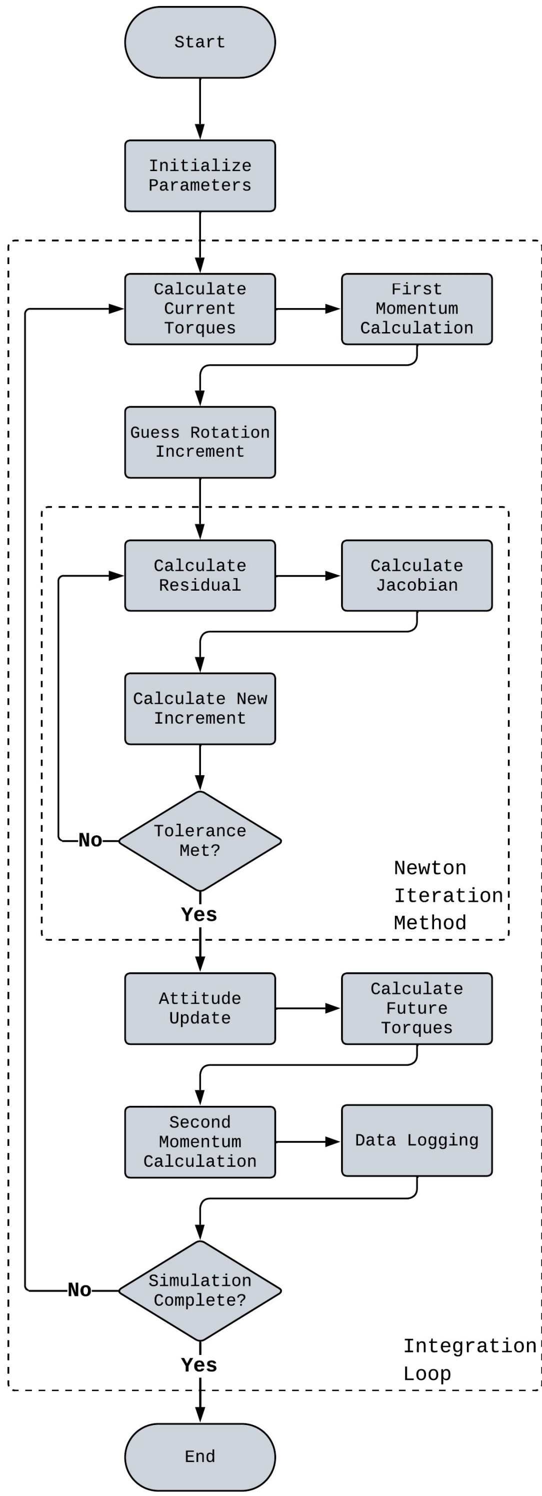

Figure 2. Architecture of an LGVI-Based MATLAB Simulation

Together, (14), (13), and (2) form the discrete equations of motion in Hamiltonian form. The Hamiltonian formulation differs from its Lagrangian counterpart by describing a system in terms of position and momentum versus position and velocity.

A vectorization operation is performed on (13) to flatten its 3×3 matrices into 3×1 column vectors. This operation translates all non-standard inertia matrices J_d back to standard J. F_k is replaced by the rotation increment vector $f_k \in \mathbb{R}^3$. F_k and f_k are related by Rodrigues' formula, which can be expressed using $c_1 = \sin\|f_k\|/\|f_k\|$ and $c_2 = (1-\cos\|f_k\|)/\|f_k\|^2$:

$$F_k = e^{S(f_k)} = I_{3\times3} + c_1 S(f_k) + c_2 S(f_k)^2. \quad (15)$$

Finally, the resulting equation can be expressed in terms of the residual $G \in \mathbb{R}^3$:

$$G = (c_1 J f_k + c_2 f_k \times J f_k) - \left(h\Pi_k + \frac{h^2}{2} M_k\right). \quad (16)$$

The residual equation forms the core of the Newton iteration. It is used to obtain the rotation increment vector f_k required for the attitude update. In (16), the Rodrigues' formula replaces the exponential function $F_k = e^{S(f_k)}$ with a trigonometric equation that can be quickly calculated by a computer.

Computational Architecture

A practical implementation of the LGVI theory into a MATLAB-based algorithm follows a Störmer-Verlet-style kick-drift-kick method, in which two half-step momentum updates frame the Lie group rotation update via a second-order splitting of the discrete Hamiltonian update map.[3] (pp. 7-8) The architecture of this algorithm is illustrated in Fig. 2.

First, initial conditions of the simulation are set before the main integration loop commences. This includes initializing the timestep, inertial matrices, hysteresis rod parameters, and attitude matrices and momentum vectors.

Second, within the main integration loop, simulation time is passed into a circular orbital propagator function that returns the satellite position r_{ECI} in the Earth-centered inertial (ECI) frame. This position is converted to the Earth-center Earth-fixed (ECEF) frame position r_{ECEF} via the transformation matrix R_{EI} by $r_{\mathrm{ECEF}} = R_{EI}(t) r_{\mathrm{ECI}}$. The latitude, longitude, and altitude values from r_{ECEF} are input into the International Geomagnetic Reference Field (IGRF) model, returning the magnetic flux density vector in local NED coordinates, B_{NED} [nT]. This flux density is rotated to ECEF and converted to Tesla, yielding B_{ECEF}. A conversion to field strength H_{ECEF} is made via $H_{\mathrm{ECEF}} = B_{\mathrm{ECEF}}/\mu_0$, where μ_0 is the magnetic permeability of free space. H_{ECEF} is rotated to the body frame via $H_B = R_{BI} R_{EI}^T H_{\mathrm{ECEF}}$, which supplies the hysteresis model. It provides the body-frame flux density $B = H_B \mu_0$ for the permanent magnet torque calculations: $\tau = m \times B$, wherein m is the magnetic dipole. Environmental torques are excluded in the present model. At each step of the integration loop, the passive magnetic torques are evaluated and applied to advance the system state. The first momentum kick is applied, calculating the next half-momentum by the following, see equation (14):

$$\Pi_{k+\frac{1}{2}} = \Pi_k + \frac{h}{2} M_k. \tag{17}$$

Third, a Newton iteration method is utilized to find the rotation increment vector f_k required for the attitude update. Before the loop begins, an initial guess for f_k using $f_{k,0} = h\left(J^{-1}\Pi_{k+\frac{1}{2}}\right)$ is made. Within the loop, the discrete residual G is calculated using (16). A Jacobian matrix $\boldsymbol{DG}$ is constructed by differentiating G with respect to f_k. The rotation increment is updated within the solver by the equation

$$f_{k,i+1} = f_{k,i} - (\boldsymbol{DG})^{-1} G. \tag{18}$$

This process is repeated until the magnitude of the residual reaches a specified tolerance, for example $tol = 1 \times 10^{-12}$ such that $\|G(f_k)\| < tol$. Then, (15) maps f_k to F_k, and the attitude update (2) is applied in the drift sequence. An analysis of the Newton iteration implemented in the simulator found that the algorithm performed exactly two iterations before f_k converged for every call.

Fourth, the future external torques are calculated by the same process described in step two, except for simulation time t_{k+1}. The future angular momentum is determined by evaluating the momentum transport equation from (14):

$$\Pi_{k+1} = F_k^T \Pi_{k+\frac{1}{2}} + \frac{h}{2} M_{k+1}. \tag{19}$$

Fifth, diagnostic data is logged for debugging. It is recommended to log variables angular velocity, pointing error, angular momentum, and orthogonality error, which are used to determine the time to steady-state attitude and to verify the physical integrity of the simulator. The integration loop continues for the duration of the simulation, propagating the satellite's attitude according to Hamilton's principle. In the absence of external torques, momentum is inherently preserved. However, the introduction of the magnetic torques of the PMAC system components cause the body's total momentum to change over time. Through the mechanism of hysteresis damping, the system's rotational energy slowly dissipates as it is converted to heat.

MAGNETIC HYSTERESIS

A CubeSat is deployed from the International Space Station with initial rotational energy that a PMAC system removes to reach and maintain steady-state attitude. Magnetic hysteresis is a process that accounts for the dissipation of rotational energy supplied by hysteresis rods, a primary component of PMAC systems. As discussed by Gerhardt in his writing on an RK45 attitude damping simulation for the Colorado Student Space Weather Experiment (CSSWE), the magnetic torques that drive hysteresis damping are produced by shifting polarities within ferromagnetic rods.[4] These shifts occur in delayed response to magnetic field changes, resulting in the transfer of rotational energy to heat.

A hysteresis loop represents the relationship between the magnetic field strength H applied to a hysteresis rod and the magnetic flux density B induced within the rod. The area of the loop determines the energy dissipation per unit volume, per cycle. It is shaped by three parameters of the rods: the coercive force H_c, the remanence B_r, and the saturation induction B_s. These parameters represent different characteristics of the hysteresis loop; saturation induction determines the maximum height of the loop, and the coercive force and remanence define the X- and Y-intercepts. Generally, higher H_c, B_r, and B_s values correlate with larger loop area, resulting in increased induced flux density and damping rates. This nonlinear behavior is modeled at each simulation step.

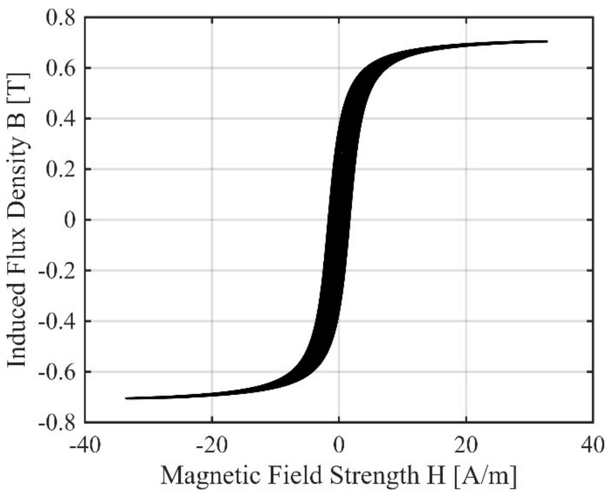

Figure 3. Reproduced Hysteresis Loop using Park et al.'s Simulation Parameters

Hysteresis models output the induced flux density B, which is used to find the rod's magnetic moment m. The torque τ exerted by the rod is found via $\tau = m \times B$. A model developed by Flatley and Henretty is widely used, which closely approximates empirically determined hysteresis data.[5] The model being nonlinear and multivalued, it is expressed in its piecewise form as

$$B = \frac{2}{\pi} B_s \tan^{-1}\left[\frac{1}{H_c}\tan(k_1)(H \pm H_c)\right], \qquad (20)$$

where the constant k varies by implementation:

$$k_1 = \left(\frac{\pi B_r}{2B_s}\right),\ k_2 = \left(\frac{\pi B}{2B_s}\right).$$

Used by Gerhardt in his simulations for CSSWE, this model was formulated by conjoining two arctangent curves fitted to hysteresis parameters, producing a multivalued result. The $\pm$ operator depends on the time derivative of the applied field strength, $\dot{H}$. A major limitation of this algebraic model is the discontinuities at the vertices of the loop that introduce artificial energy changes.

A more accurate representation is the differential version used by Park et al. in their LGVI attitude damping simulations for RAX.[6] In this method, B becomes a state variable integrated by the solver and is calculated using *sin* and *cosine* functions that are numerically more stable than arctangent:

$$\dot{B} = \frac{2B_s}{H_r \pi}\left(\frac{(H \pm H_c)\cos(k_2) \mp H_r \sin(k_2)}{2H_c}\right)^2 \dot{H}. \qquad (21)$$

(21) provides a continuous model compatible with the fixed-step LGVI and is implemented through a simple forward Euler integration scheme. The Euler update outputs B for the next timestep and can be expressed as

$$B_{k+1} = B_k + h \cdot \dot{B}(H_k, \dot{H}_k, B_k). \qquad (22)$$

This differential model treats the magnetic induction of the rod as a dynamic state and provides a more physically accurate energy dissipation effect.

SIMULATOR VALIDATION

Two simulation frameworks are developed in MATLAB for the purpose of evaluating the performance of proposed PMAC system configurations and informing the selection of the dimensional sizing for GASRATS. The frameworks follow the architecture of the CSSWE and RAX simulations. The development follows a three-step validation chain. This process is necessary for ensuring the fidelity of new simulators, and it resulted in two validated pairs of attitude integrators coupled with hysteresis models. The baseline framework incorporates MATLAB's ode45 solver with a piecewise, algebraic Flatley-Henretty hysteresis model, based on the simulation architecture discussed by Gerhardt. It utilizes a stable quaternion formulation that avoids trigonometric singularities (e.g., gimbal lock) present in Gerhardt's Euler-angle kinematic equations. The second simulation framework incorporates a second-order LGVI with a differential Flatley-Henretty hysteresis model, following the architecture proposed by Park et al.

Reference Validation

The developed baseline RK45 and LGVI simulation frameworks are validated against the literature to test the fidelity of the encoded physics by comparing convergence behavior. Both frameworks demonstrate strong agreement with their respective benchmarks. The angular velocity damping profile of the RK45 framework reproduces key features of Gerhardt's results. A shoulder region of the reference angular velocity reflects similar structure and location. The expected consolidation of angular velocity components into individual strands and subsequent divergence and reconvergence is evident. Additionally, the time to steady-state attitude in the RK45 framework is in close agreement with the literature. For the LGVI framework, the angular velocity damping profile and evolving shape of the pointing error align with Park et al.'s results, see Fig. 4. The settling times are comparable, with a variance of one simulated orbit (approximately 1.64 hours). Overall, the validation confirms a satisfactory alignment between the present frameworks and references, qualifying for integration with GASRATS parameters.

Cross-Framework Verification

Having been validated against their respective source literature, a cross-framework verification study is conducted to assess the consistency of damping behavior between simulation frameworks. Identical convergence patterns are not expected because the architectures are different. However, this study is valuable for discovering sensitivities associated with each implementation. Simulation parameters for GASRATS are used (see Table 1).

The hysteresis parameters for GASRATS are determined by comparison to Gerhardt's hysteresis parameters. The same apparent coercive force and apparent remanence are adopted, as GASRATS uses the same rod alloy, HyMu-80. B_s' was calculated by a complicated procedure provided by Gerhardt that is beyond the scope of this paper. Apparent hysteresis parameters are implemented because they account for the demagnetizing effect of the rod's own magnetic field. While this verification step employs apparent properties, the previous LGVI validation with Park et al.'s Scenario 4 uses intrinsic properties. Therefore, the resulting angular velocity profile in Fig. 4 represents accelerated damping from fully saturated rods (see also Fig. 3).

The baseline RK45 and LGVI simulations are configured to a maximum timestep of 0.05 seconds. Also, the LGVI simulation implements the 14$^{\text{th}}$ generation of the International Geomagnetic Reference Field (IGRF-14), which is precomputed at a coarse interval of one second to save computation time.

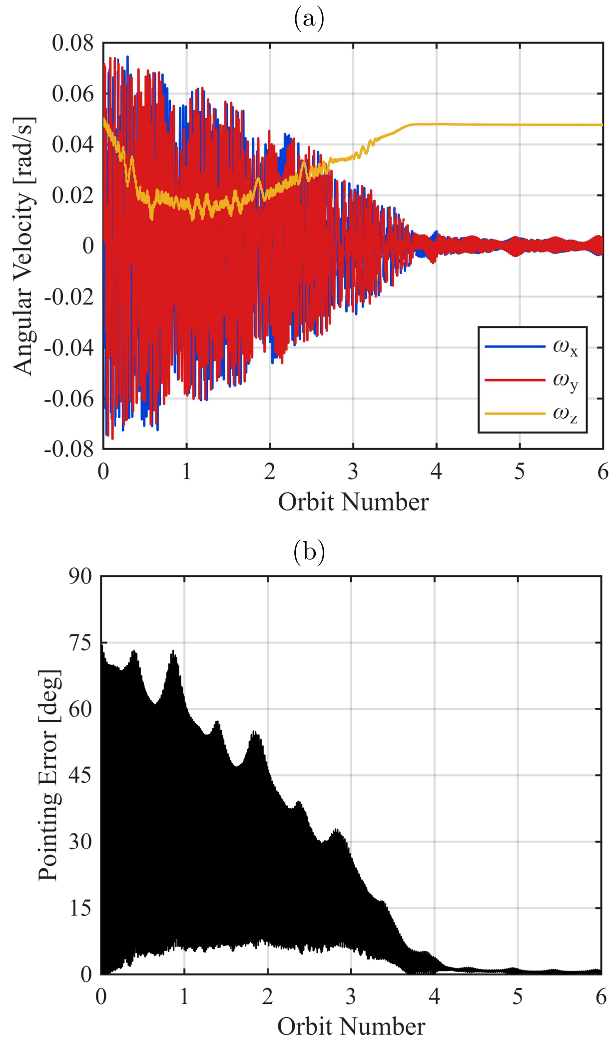

Figure 4. Reproduced LGVI Simulation Results of Park et al.'s Scenario 4: (a) Angular Velocity, (b) Pointing Error

The two simulation frameworks are evaluated based on the consistency of qualitative features and the time to steady-state attitude (see Fig. 5). The steady-state criteria for GASRATS is driven by the mission payload, a transparent patch antenna. Patch antennas possess narrower bandwidth than omnidirectional antennas, and a more stringent pointing-error requirement may be needed to ensure link closure during ground station passes. While Gerhardt's simulations were evaluated using a steady-state criterion of holding pointing error $\beta < 15°$, the steady-state requirements for this study are defined as holding $\|\omega\| < 0.5$ deg/s and $\beta < 10°$.

Table 1. Initial GASRATS Parameters for Cross-Framework Verification		
Parameter	**Symbol**	**Value**
Altitude [km]	h	400
Inclination [deg]	i	51.6
Initial Angular Velocity [deg/s]	ω_0	[10, 5, 5]
Initial Orientation	q_0	[1, 0, 0, 0]
Mass [kg]	m	2
Transverse Inertia [$\text{kg} \cdot \text{m}^2$]	J_{xx}, J_{yy}	0.0093
Axial Inertia [$\text{kg} \cdot \text{m}^2$]	J_{zz}	0.0033
Permanent Magnet Strength [$\text{A} \cdot \text{m}^2$]	m_{pm}	0.3, 0.03
Number of Rods	n	4
Total Rod Volume [m^3]	V	5.58×10^{-7}
Apparent Coercive Force [A/m]	H_c'	12
Apparent Remanence [T]	B_r'	0.004
Apparent Saturation Induction [T]	B_s'	0.0128

The resulting profiles of the angular velocity and pointing error displayed in Fig. 5 demonstrate attitude convergence but do not exhibit rigorous qualitative agreement. This is expected and can be attributed to the architectural differences between frameworks; the two simulators employ different magnetic field and hysteresis models. Additionally, the LGVI is configured with candidate magnet strength 0.03 $\text{A} \cdot \text{m}^2$ as the 0.3 $\text{A} \cdot \text{m}^2$ configuration exhibited oscillatory nonconvergent behavior for Table 1 parameters. The RK45 simulation reaches steady-state attitude in 7.29 days, while the LGVI simulation reports a steady-state convergence in 12.42 days. This delayed convergence is caused by the bounded oscillatory pattern of the LGVI simulation's pointing error after day three that marginally exceeds the 10° threshold. Thus, it is determined that this pointing error threshold is overly stringent, and a 15° requirement consistent with Gerhardt's methodology is adopted going forward.

Invariant Preservation Analysis

The final validation step is an invariant preservation analysis that tests the conservation of kinetic energy, angular momentum, and structure orthogonality to ultimately determine which simulation framework is most physically realistic. The test configuration adopts a torque-free adaptation of the simulation frameworks used in the cross-framework validation. A simulation duration of 50 orbits or 3.21 days is implemented.

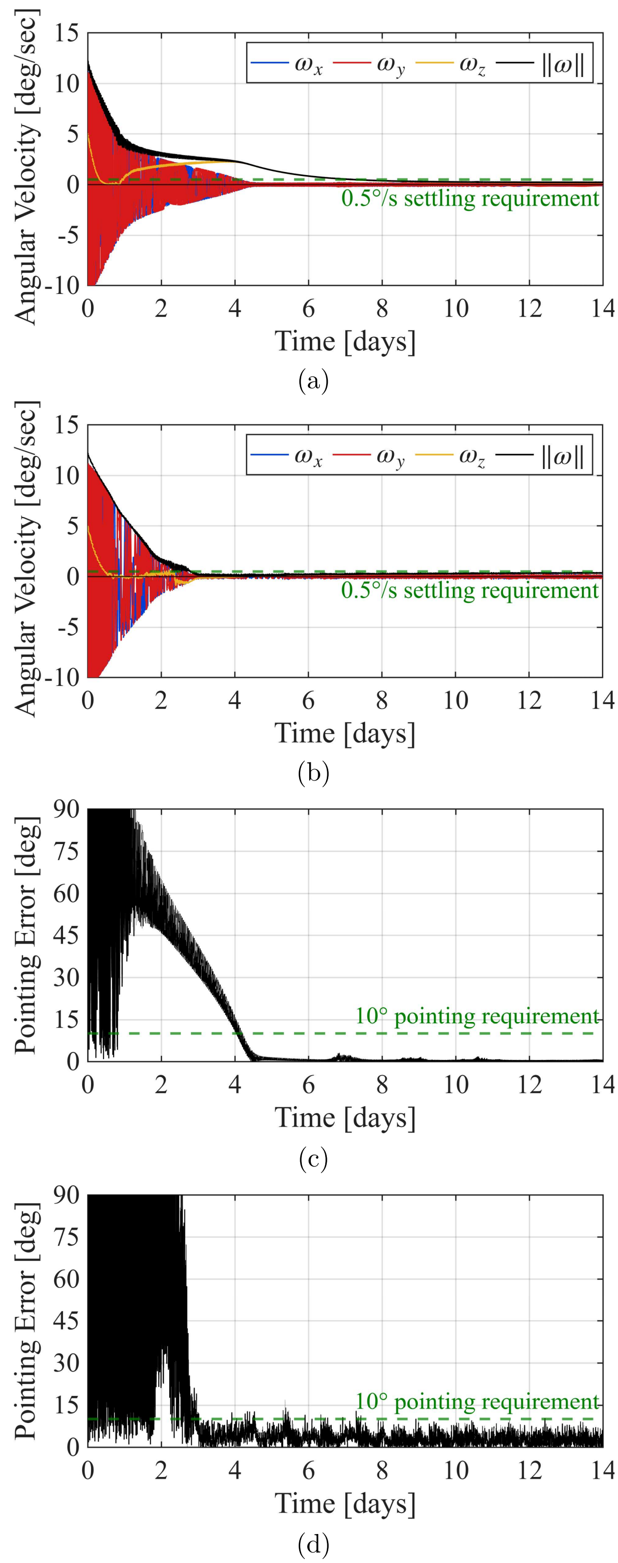

Figure 5. Cross-Framework Verification: (a) RK45 Angular Velocity, (b) LGVI Angular Velocity, (c) RK45 Pointing Error, (d) LGVI Pointing Error

The test reveals that both the RK45 and LGVI frameworks achieve a high degree of fidelity in invariant preservation, and the RK45 framework outperforms initial expectations. This is attributed to the RK45 framework's restricted maximum timestep of 0.05 seconds that corresponds with the LGVI's fixed timestep. Both models exhibit excellent structure preservation, approaching relative orthogonality and norm errors on the order of 10^{-12}. While the LGVI demonstrates superior angular momentum conservation with a relative error on the order of 10^{-13}, the RK45 integrator approaches a relative momentum error on the order of 10^{-9}. This figure is expected to decline further for increasing simulation timespans. Although the LGVI shows lower kinetic energy (KE) preservation for the tested duration, it more accurately preserves angular momentum to a significant degree while maintaining rotation-matrix orthogonality of 10^{-11} relative error. Therefore, the LGVI is selected for the Monte Carlo analysis, with KE error retained as an implementation limitation. Additionally, the torque-free LGVI performs the 50-orbit test in 51.3 seconds—twice as fast as the RK45 integrator.

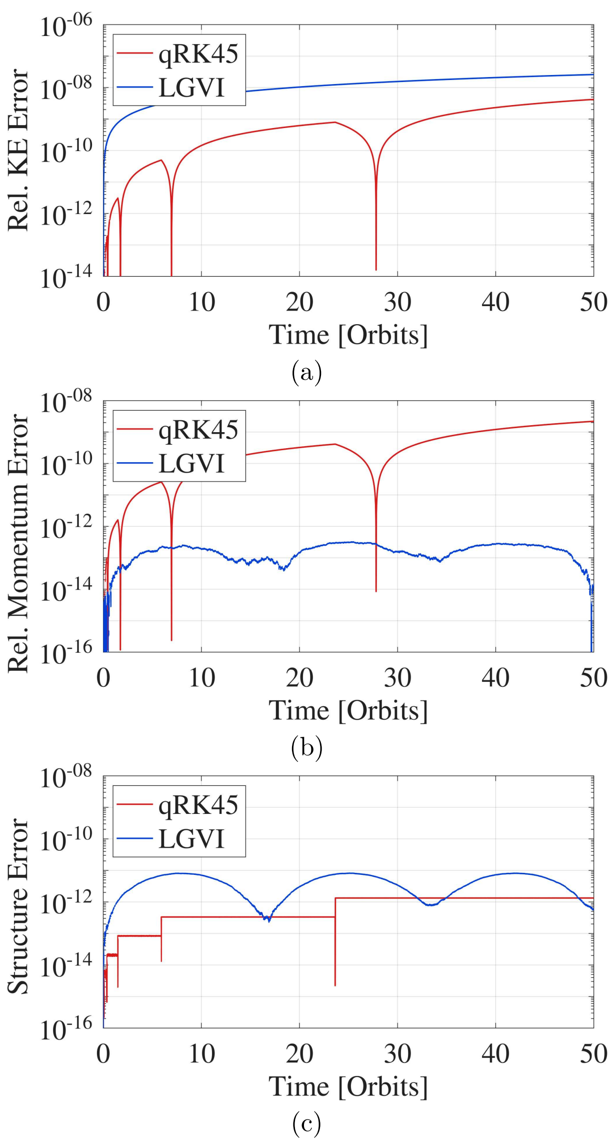

Figure 6. Invariant Preservation Analysis Results: (a) Relative Kinetic Energy Error, (b) Relative Momentum Error, (c) Orthogonality/Norm Error

MONTE CARLO ANALYSIS

A 1,001-trial Monte Carlo analysis is conducted on GASRATS mission parameters to inform key mission design decisions such as the hysteresis rod count and satellite size. This comprehensive test utilizes the validated LGVI simulation framework discussed in previous sections, which is selected for its angular momentum preservation (see Fig. 6) and computational efficiency. As preliminary testing indicates the inadequate hysteresis damping provided by two rods, the computational power available for the test is focused on configurations with four rods (801 trials versus 200).

The trial sets are split quarterly towards each CubeSat configuration (1U, 1.5U, 2U, 3U). The additional four-rod trial is a control trial using the same parameters detailed in the cross-framework verification study (see Table 1) for verifying batch integrity. MATLAB's Parallel Computing Toolbox was used to distribute the Monte Carlo trials across eight CPU workers on a 13th Gen Intel® Core™ i7-13700K CPU. Initial angular velocity magnitudes range from zero to 35°/s, whose upper bound represents the nominal deployment velocity limit $\|(5°/s\,, 5°/s, 5°/s)\|$ from the Nanoracks CubeSat Deployer multiplied by a safety factor of four.

Preliminary testing reveals that an angular velocity asymptote at $\|\omega\| = 0.5°/s$ exists for certain cases, causing false non-convergence. To ensure accurate convergence reporting, the settling requirement of $\|\omega\| < 0.5°/s$ is raised to $\|\omega\| < 0.75°/s$. Each trial is run for 30 simulated days, and a trial is categorized as successful if the steady-state criteria of $\|\omega\| < 0.75°/\text{s}$ and $\beta < 15°$ is reached within the mission requirement of seven days and maintained until the end of the simulation.

CONCLUSIONS

The results of the analysis (see Fig. 7 and Table 2) confirm expectations for configurations with two rods to produce insufficient damping. Across the initial angular velocity range of 0 to 35°/s, 50% of trials for two rods reach steady-state attitude within the mission requirement, while 89% of trials for four rods reach steady-state within the mission requirement. The 1.5U configuration demonstrates a 100% success rate for settling within mission requirements for initial angular velocities up to three times the nominal velocity limit. These results inform the final recommendation of a four-rod PMAC system and 1.5U size selection for GASRATS.

Based on the results, it is determined that higher initial rotational kinetic energy corresponds with longer damping times. As initial rotational kinetic energy scales with velocity and inertia, it is expected for the settling duration to increase as initial angular velocity and inertia increase. This trend is realized for sizes 1.5U, 2U, and 3U, whereas the 1U case is an anomaly. Although the 1U was expected to consistently reach steady-state within seven days due to its low inertia, this configuration demonstrates the poorest damping success rate out of the four-rod trials. This phenomenon can be attributed to the 1U's inertial symmetry.

Table 2. 4-Rod Monte Carlo Results

Configuration / Bin	Pass Rate	Percentage
GASRATS 1U / 4-Rod Configuration		
Bin 1: $00.00°/s \le \|\omega\| < 08.66°/s$	51/56	91.07%
Bin 2: $08.66°/s \le \|\omega\| < 17.32°/s$	30/39	76.92%
Bin 3: $17.32°/s \le \|\omega\| < 25.98°/s$	29/54	53.70%
Bin 4: $25.98°/s \le \|\omega\| < 35.00°/s$	25/51	49.02%
GASRATS 1.5U / 4-Rod Configuration		
Bin 1: $00.00°/s \le \|\omega\| < 08.66°/s$	49/49	100.00%
Bin 2: $08.66°/s \le \|\omega\| < 17.32°/s$	51/51	100.00%
Bin 3: $17.32°/s \le \|\omega\| < 25.98°/s$	46/46	100.00%
Bin 4: $25.98°/s \le \|\omega\| < 35.00°/s$	48/54	88.89%
GASRATS 2U / 4-Rod Configuration		
Bin 1: $00.00°/s \le \|\omega\| < 08.66°/s$	53/53	100.00%
Bin 2: $08.66°/s \le \|\omega\| < 17.32°/s$	51/51	100.00%
Bin 3: $17.32°/s \le \|\omega\| < 25.98°/s$	44/47	93.62%
Bin 4: $25.98°/s \le \|\omega\| < 35.00°/s$	37/49	75.51%
GASRATS 3U / 4-Rod Configuration		
Bin 1: $00.00°/s \le \|\omega\| < 08.66°/s$	54/54	100.00%
Bin 2: $08.66°/s \le \|\omega\| < 17.32°/s$	51/51	100.00%
Bin 3: $17.32°/s \le \|\omega\| < 25.98°/s$	37/44	84.09%
Bin 4: $25.98°/s \le \|\omega\| < 35.00°/s$	34/52	65.38%

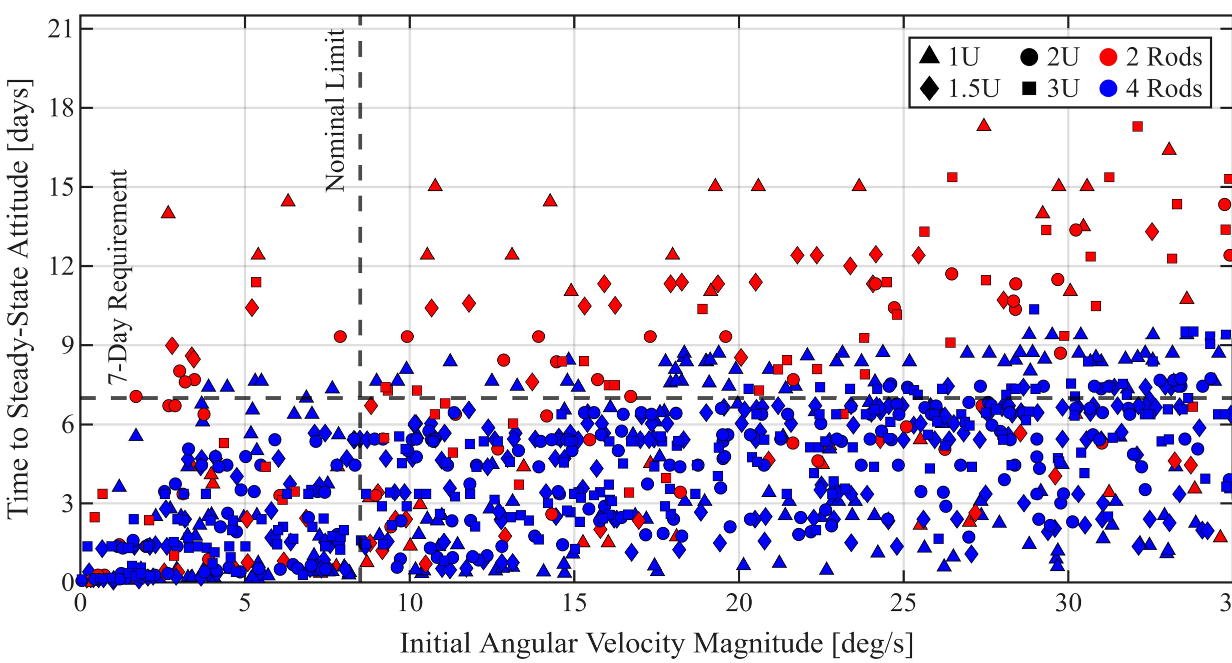

Figure 7. 1,001-Trial Monte Carlo Analysis for GASRATS

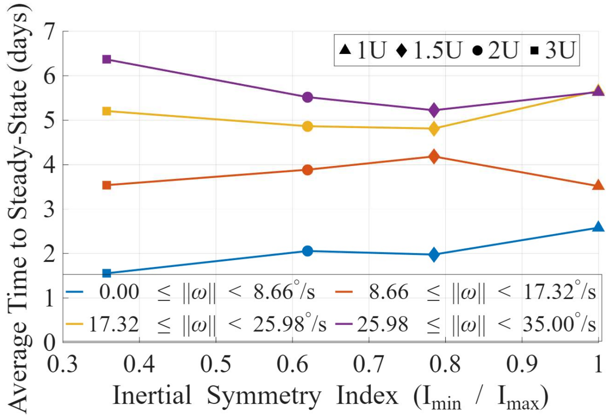

Figure 8. Inertial Symmetry vs. Settling Time for 4-Rod Trials

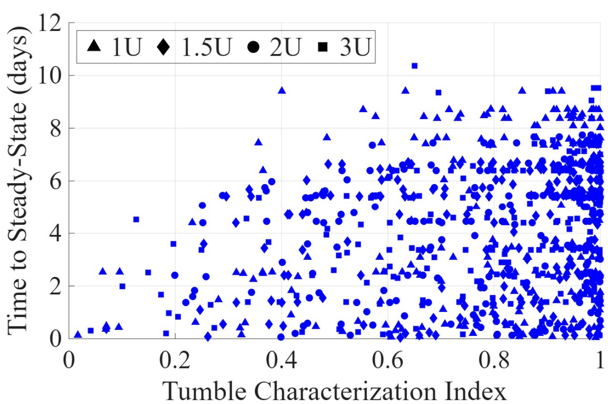

Figure 9. Tumble Characterization Index vs. Settling Time for 4-Rod Trials

Index 0 indicates a pure Z-axis spin, and index 1 indicates a pure X-Y tumble.

A rotating body with inertial symmetry experiences limited rotational energy transfer between body axes. Hence, 1U configurations respond poorly to non-ideal deployment scenarios such as end-over-end tumbles wherein angular velocity components on the X- and Y-axes dominate the Z-component velocity. As an arrangement of hysteresis rods on the X- and Y-axes primarily removes rotational energy from the Z-axis, an end-over-end tumble for a CubeSat with inertial symmetry realizes less efficient damping. Fig. 8 and Fig. 9 represent the effect of inertial symmetry and tumble type on settling time. Notice in Fig. 8 that for average nominal deployment velocities, the 1U configuration possesses the longest average damping time despite having the least inertia. Conclusions of the Monte Carlo study are limited to comparative analysis between configurations and do not provide reliable estimates for damping times in LEO.

FUTURE WORK

Dynamic environmental torques will be implemented in the simulator to more accurately model the space environment. Beyond Euclidean spaces, LGVIs can be formulated on relativistic manifolds, such as Minkowski spacetime and the Poincaré group, for simulators that preserve relativistic effects such as Thomas precession and time dilation.

ACKNOWLEDGEMENTS

The author thanks Larry Catalasan for insightful discussions on advanced dynamics, and Dr. Owen Abbe for his support and for fostering an early interest in space.

Code Availability:

github.com/TrumanTAbbe/CubeSat-LGVI-Magnetic-Attitude-Control-Simulators

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